

Constitutional LLM Analysis Final Progress Report

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1. Completed Tasks

I have completed all the tasks we explicitly discussed, verified by the code and its output (5_validation.ipynb):

- Updated Open-Ended Questions: I've fully updated the open-ended questions (HOSTERM, HOGTERM, CONTERM, CHFTERM, HOSNAME) to align with the new instructions, removing the simple '1' and '99' options where appropriate and ensuring the LLM only answers based on the provided instructions.
- Validation Pipeline and Codebook: I've updated my validation pipeline to generate a detailed Excel report (Validation_Detailed_Report.xlsx). This file now includes a separate sheet, the Codebook_Options, which contains all questions, their possible answers (0, 1, 96, 98, 99, etc.), and the instructions, as requested for law readers.
- Additional Scoring Metrics: The system already reports F1, Precision, and Recall, along with Strict and Relaxed scoring metrics (treating 96–99 as 0), as requested.

2. Current Accuracy

The current system, using the updated few-shot prompting and hierarchical processing, shows the following performance metrics based on the strict evaluation (exact match) and relaxed evaluation (96-99 treated as 0):

Category	Strict Accuracy (Exact Match)	Relaxed Accuracy (96-99=0)
Root Questions	71.27%	74.18%
Conditional: Answering (Truth != 99)	60.00%	66.67%
Conditional: Skipping (Truth == 99)	51.58%	92.63%
Global	63.22%	73.01%

This skipping determinant assumes that if the question option picked was 99, then it was skipped, whether that be on its own volition otherwise or because it detected that a root conditional was not met. Otherwise, the validation system treats the question answer as if the LLM did not skip and legitimately answered the question.

Example of not skipping because a conditional:

Country	Type	Variable	Type	ID	Question	Condition	H	Human Text	A	AI Text
United States	1791	AMEND	Root	v70	Does the constitution provide for at least one procedure for amending the constitution?		1	1 (Yes)	1	1 (Yes)
...	...	...	...	...	...	...	...	...	...	...
United States	1791	AMNDAPPR_7	Cond	v76	Who approves amendments to the constitution?	(Asked only if AMEND is answered 1)	1	1 (Head of State (use this choice for single executive systems))	1	1 (Head of State (use this choice for single executive systems))

Example of skipping because a conditional:

Country	Type	Variable	Type	ID	Question	Condition	H	Human Text	A	AI Text
Taiwan	1947	JUDCRTS_2	Root	v309	For which of the following specialized courts does the constitution contain provisions?		0	0 (No)	0	0 (No)
...	...	...	...	...	...	...	...	...	...	...
Taiwan	1947	CONLIM	Cond	v356	What restrictions are in place regarding the number of terms judges on the Constitutional Court may serve?	(Asked only if JUDCRTS is answered 2)	99	99 (Not Applicable)	99	99 (Not Applicable)

The significant gain in "Conditional: Skipping" accuracy under the relaxed score (from 51.58% to 92.63%) suggests that a major source of disagreement is the model choosing '0' (No) when the human coder used a "Not Applicable" or similar code (96, 97, 98, or 99).

3. Suggestions for Accuracy Improvement

Given the flexibility of the current LLM prompting system (as long as the outputted format is consistent), accuracy could be significantly improved by the following methods:

- **Multi-Model Verification:** Implementing a second system (e.g., using a different LLM or model) to re-run the answers and double-check the results of the first model.
- **Enhanced Chain-of-Thought Prompting:** Further refining the prompt's CoT structure to force the LLM to think harder about its reasoning and clearly cite the constitutional text before providing a final answer.

- **Critical Thinking Examples (A Key Request):** Providing more high-quality, non-production examples in the prompt to guide the LLM's critical thinking. I currently only use a handful of generic examples (provided in Section 5) and I am unsure if providing the actual answers from the validation data is allowed. If the law team could provide 5-10 mock examples of nuanced coding decisions (e.g., complex condition, ambiguous language) that would be very helpful.

These are my thoughts on the current state of the system. While the relaxed metrics are encouraging, these larger-scale modifications and agent integrations would lead to a much more accurate and robust system.

4. Additional Results

a. Global Metrics (All Questions)

This table shows the overall performance across all coded variables.

Metric	Global Strict (Exact Match)	Global Relaxed (96-99 treated as 0)
Accuracy	63.22%	73.01%
F1 Score	0.6255	0.7156
Precision	0.6525	0.7267
Recall	0.6322	0.7301

b. Hierarchical Breakdown (Strict Scoring)

Strict scoring requires an exact match for the predicted code (e.g., predicted '98' must match a true '98').

Question Category	Accuracy	F1 Score	Precision	Recall
Root Questions	71.27%	0.7212	0.7884	0.7127
Conditional: Answering (Truth ≠ 99)	60.00%	0.6267	0.6764	0.6000
Conditional: Skipping (Truth = 99)	51.58%	0.6806	1.0000	0.5158

c. Hierarchical Breakdown (Relaxed Scoring)

Relaxed scoring is more lenient, treating all "Not Specified" or "Not Applicable" codes (96, 97, 98, 99) as a single value (0) for comparison.

Question Category	Accuracy	F1 Score	Precision	Recall
Root Questions	74.18%	0.7405	0.7818	0.7418
Conditional: Answering (Truth ≠ 99)	66.67%	0.6426	0.6582	0.6667
Conditional: Skipping (Truth = 99)	92.63%	0.9617	1.0000	0.9263

5. Interpretation of Metrics

Metric	Definition	Interpretation
Precision	Out of all the codes the model predicted for a category, how many were correct?	A High Precision (like 1.0000 for Strict Skipping) means when the model predicts the correct answer/skip for that category, it is almost always right.
Recall	Out of all the questions that should have been coded correctly in a category, how many did the model capture?	A High Recall means the model is good at finding and answering (or skipping) most of the questions that belong in that category.
F1 Score	The harmonic meaning of Precision and Recall.	This is the overall, balanced measure of performance. It is useful when the dataset is imbalanced (e.g., more "Skip" cases than "Answer" cases). The F1 score is generally considered the best single number to compare the overall strength of two classification models.

5. Questions / Answer Examples Provided In Prompt

Below are all of the questions within the prompt system. This can easily be expanded and re-run for many more questions at the cost of tokens. Only three were provided in prompt order to minimize token usage and run time, as many runs were undertaken to get consistent results.

EXAMPLES (Follow this logic strictly)

Example 1: Explicit Right

Text: "All citizens shall have the right to free speech."

Question: [v12] Freedom of speech?

1. Yes 2. No 98. Not Specified

ANALYSIS: The text in Article 3 explicitly grants the right to free speech.

FINAL: v12|1

Example 2: Explicit Denial

Text: "The President may not dissolve Parliament."

Question: [v55] Can the Executive dissolve the Legislature?

1. Yes 2. No 98. Not Specified

ANALYSIS: The text explicitly forbids the President from dissolving Parliament.

FINAL: v55|2

Example 3: Silence (Not Specified)

Text: "The President is elected for 5 years."

Question: [v99] Is there a Vice President?

1. Yes 2. No 98. Not Specified

ANALYSIS: The text mentions the President but is completely silent on the office of a Vice President. I will not infer one exists.

FINAL: v99|98